Andrea Valente

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Authorised to work in Spain · Open to relocation to Valencia

Summary

AI Computer Vision Engineer working across model development, geometric vision and inference systems. Builds evaluation pipelines and practical tools that turn research findings into client decisions and operational R&D workflows.

Experience

Deep Vision Consulting — Modena, Italy

AI Computer Vision Engineer — Apr 2024 – Present

- **Package-seal inspection:** Became primary implementation owner of a Python analysis suite for a global packaging manufacturer, combining image segmentation, geometric measurements and force features.
- Designed thin-plate-spline defect synthesis and a tailored U-Net workflow, using source-aware splits and reflection augmentation to address scarce examples and misleading edge reflections.
- Built the cloud-integrated dashboard used by the sealing team on new samples as an operational R&D tool, with queued GPU tasks and failure inspection; supported client engineers with deployment and handover.
- **Efficient pose estimation:** Implemented Python training, inference, camera correction, evaluation and tracking for 6DoF head pose; comparisons supported joint selection of a 0.6 MB LCNet model versus a 2.4 MB reference, with equivalence within client-agreed rotation/translation thresholds.
- Integrated ONNX Runtime with WebGPU, graph capture and GPU preprocessing into a pre-production test application; profiled runtime behaviour and tested across high-, mid- and low-tier devices.
- **3D reconstruction:** Created the client's first quantitative reconstruction framework for a global luxury eyewear group, annotating and fitting 68 face scans to FLAME and defining region-specific mesh errors and eyewear-relevant measures.
- Designed the acquisition protocol and prepared a 48-subject, 7,608-take benchmark; findings on 3D within-take stability and 2D across-session reproducibility informed a joint client/team decision to prioritise more viable facial-measurement approaches.
- **Point-cloud geometry:** Developed Python RANSAC-based fitting and synthetic-data tools for smart-lighting point clouds; the package was evaluated on selected shapes through visual feedback.

AI Computer Vision Engineer (Intern) — Sep 2023 – Mar 2024

- Evaluated NeRF/MVS and deformable meshes against virtual try-on acquisition and computation constraints; generated approximately 60,000 synthetic face images and trained a shape-regression network, identifying scale ambiguity and synthetic-to-real limitations.

Vaia Innovations — Remote, part-time

Frontend Web Developer — Jun 2022 – Sep 2022

- Built responsive interfaces from Figma designs for article exploration and writing on a Web3 publishing platform, using TypeScript, Vue.js and Tailwind CSS.

Skills

ML and evaluation: Python, PyTorch, OpenCV, NumPy, MLflow; synthetic data, segmentation, model training and source-aware validation.
Geometric vision: 3D reconstruction, FLAME mesh fitting, camera geometry, point-cloud processing, RANSAC, SciPy optimisation, Open3D, PyTorch3D, Blender.
Inference and testing: ONNX Runtime, WebGPU, TFLite, GPU profiling, Perfetto, Playwright, cross-device testing.
Software and delivery: Git, Docker, Celery, RabbitMQ, PostgreSQL, cloud-storage integration; TypeScript, Vue.js, Tailwind CSS.

Education

MSc in Artificial Intelligence — University of Bologna
110/110 with honours — Sep 2021 – Mar 2024

BSc in Computer Science — University of Bologna
110/110 with honours — Sep 2018 – Dec 2021

Languages

Italian: native. English: professional working fluency; MSc taught in English.
IELTS Academic: overall band 7.0 (Jul 2021).